

WASP-52b

R-1751

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_200915 · created 2026-08-02T19:52:55+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459107.7768 +/- 0.0057 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.168 +/- 0.062
Transit depth (Rp/Rs)^2	2.82 +/- 2.07 [%]
Orbital Inclination (inc)	81.97 +/- 2.69
Airmass coefficient 1 (a1)	0.045 +/- 0.034
Airmass coefficient 2 (a2)	2.87 +/- 1.51
Scatter in the residuals of the lightcurve fit is	82.4 %
Best Comparison Star	None
Optimal Aperture	0.99
Optimal Annulus	3.0
Transit Duration (day)	0.037 +/- 0.031

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.168 $\pm$ 0.062 vs 0.1646 (deviation 0.04 σ)
Duration fit vs published	0.037 d vs 0.0758 d (deviation 1.0 σ)
Mid-time O–C	13.9 min
Depth SNR	2.2
Residual scatter	82.4 %

Light curve

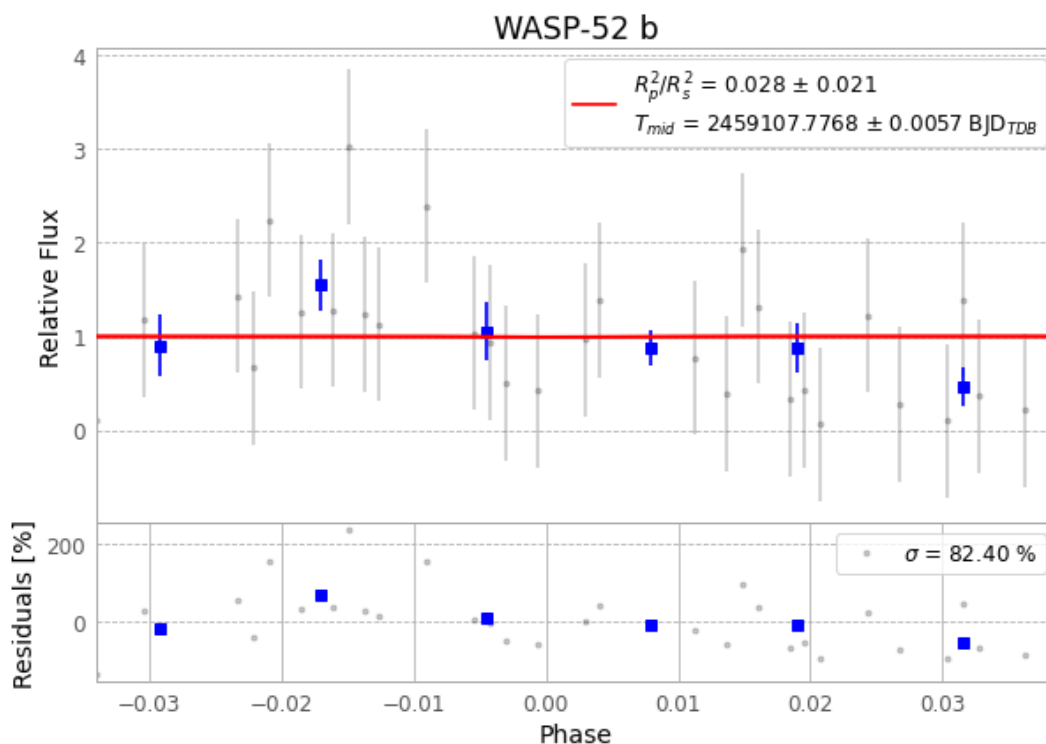


Figure 1 — FinalLightCurve_WASP-52 b_2020-09-14.png (SHA-256 8ec6deb5058a06d6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [509, 101], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 bc3546584b2f8e16...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

67 FITS frames (44 MB), WASP-52200915045713.FITS → WASP-52200915081512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-200915.log (1f766bf32597...), FinalLightCurve_WASP-52 b_2020-09-14.png (8ec6deb5058a...), FinalLightCurve_WASP-52 b_2020-09-14.csv (01007831cee9...)

Compute resources

Wall time 140.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 19:52Z.